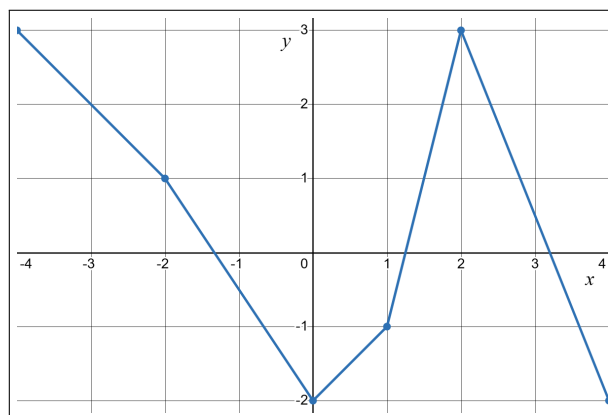


1. (Fall 2024 #6) Consider the function: $f(x) = 2x^{5/3} + 5x^{2/3}$.

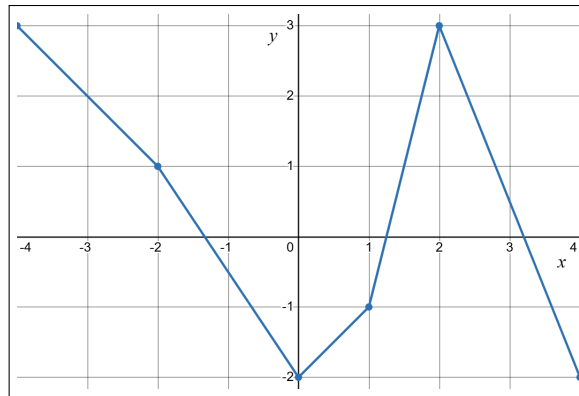
(a) Find the x -values of all local maxima and local minima of $f(x)$ on its domain.

(b) Find the absolute maximum value of $f(x)$ on the interval $[-1, 1]$.

2. (Fall 2020, #7) Consider the following graph:



- (a) Suppose that this is the graph of a function $f = f(x)$. In particular, $f(-1) = -1/2$.
- i. Consider the function $g(x) = f(3x - 4)$. Compute $g'(2)$, if possible. If this is not possible, then explain why.
 - ii. Consider the function $h(x) = f(f(x)) + \ln(1 - \tan(\pi f(x)/2))$. Compute $h'(-1)$, if possible. If this is not possible, then explain why.



(b) Now suppose that this is the graph of the **derivative** u' of a function $u = u(x)$, and suppose moreover that $u(-2) = 0$.

i. Compute an approximate value of $u(-2.1)$. Show your work.

ii. For each of the following statements, indicate whether it is true, false, or you need more information to make the conclusion, and give a brief explanation.

u is concave up on $(0, 2)$ **True** **False** **Need More Info**

u has a local maximum on $[2, 4]$ **True** **False** **Need More Info**

3. (Fall 2020, #10) A clothing store produces and sells suits. The total (cumulative) cost, in dollars, of producing q suits is

$$C(q) = 4000 + 0.25q^2$$

To sell q suits, the price per suit, in dollars, must be

$$p = 150 - 0.5q$$

- (a) What is the profit from selling q suits?
- (b) How many suits must the store produce and sell to maximize profit?
- (c) What is this maximum profit?
- (d) What price per suit must the store charge to maximize profit?

4. (Fall 2019, #1) Consider the function $f(x) = 3x^{\frac{5}{3}} + 60x^{\frac{2}{3}}$.

(a) Show that $f'(x) = \frac{5(x+8)}{\sqrt[3]{x}}$. Then find and classify all local extrema of f .

(b) Find the values of the global extrema of f on the domain $[-1, 1]$.

(c) Justify briefly, but precisely, why f has global extrema on the interval $[-1, 1]$.

5. (Fall 2017, #7) Find all points in the interval $[-2, 2]$ on which the function

$$f(x) = \frac{x^{\frac{2}{3}}}{2 + x^2}$$

has a local maximum or minimum, and find the absolute maximum and minimum values of f on this interval.